

TEST BANK

CHAPTER 11

HUMAN IMPACTS ON THE SEA

Matching

In this section, you are to select from the choices listed in column B the term or phrase that best matches the term or phrase listed in column A and then to place the letter for the selected choice from column B in the space provided to the left of the corresponding item in column A.

[Note to instructors: Here, the choices are correctly matched and thus should be randomly assorted before the exam is prepared.]

Column A		Column B
___ 1. IronEx	—————>	A. Moss Landing Marine Lab experiment
___ 2. demersal	—————>	B. bottom dwelling
___ 3. guano	—————>	C. natural fertilizer
___ 4. curies	—————>	D. units of measure for radioactivity
___ 5. T.H. Huxley	—————>	E. said that the world's major fisheries are inexhaustible
___ 6. krill	—————>	F. euphausiid shrimp
___ 7. Minamata disease	—————>	G. caused by methyl mercury
___ 8. eutrophication	—————>	H. excess nutrients
___ 9. Greenhouse gases	—————>	I. CO ₂ , CH ₄ , and CFCs
___ 10. Garrett Hardin	—————>	J. author of <i>Tragedy of the Commons</i>

Multiple Choice

Circle the choice that best completes or answers the given questions.

[N.B.: The correct answer to each of the questions is indicated by an asterisk. For most of the questions, the letter "a" is the correct choice; reorder the choices as you see fit.]

- Herring, anchovies, and anchovetas are commonly known as _____.
 - clupeoids *
 - gadoids
 - demersals
 - mackerel
 - whales
- Most marine biologists believe that the character Doc in John Steinbeck's novel *Cannery Row* was probably based on which of the following individuals?
 - Ed Ricketts *
 - John Steinbeck
 - James Nybakken
 - Jacques-Yves Cousteau
 - (Instructor, place your name here)

182 Test Bank for Marine Biology

3. How many Exxon captains does it take to pilot a tanker through the inner passage of Alaska?
 - a. One with a fifth
 - b. One, plus two more to mix the drinks
 - c. Only one half of a captain, because he sees double
 - d. Any of the above *
4. Dumping mercury into a bay along the coast of Japan led to the permanent neurological disorder called _____.
 - a. Minamata disease *
 - b. chloracne
 - c. eutrophication
 - d. saponin
 - e. gadoids
5. A hypothetical radioisotope with a half-life of 22 years is to be disposed of in a storage facility. How many years must pass before this material is considered safe?
 - a. 2.2 years
 - b. 22.0 years
 - c. 330 years *
 - d. 2,200 years
 - e. 23,000 years
6. Which one of the following chemicals is a natural shark repellent?
 - a. Saponin *
 - b. Guano
 - c. Palytoxin
 - d. Clupeoid
 - e. Gadoid
7. Which of the following indicates the largest number of fishes that can be harvested on a continuing basis without depleting the stocks?
 - a. Gross tonnage
 - b. Net tonnage
 - c. Greatest yield
 - d. Maximum yield
 - e. Maximum sustainable yield *
8. In attempting to track the total amount of carbon (as CO₂) added to the atmosphere each year, scientists are using which of the following as a distinctive tag?
 - a. O₃
 - b. O₂
 - c. O₃:O₂
 - d. Pb
 - e. ¹³C:¹²C *
9. Humans dump ____ into the world's oceans.
 - a. household trash
 - b. radioactive wastes
 - c. industrial wastes
 - d. sewage
 - e. All of the above *
10. The span of the current standard fishing and exclusive economic zone for all coastal nations is which of the following?
 - a. 200 miles *
 - b. 200 kilometers
 - c. 50 miles
 - d. 12 miles
 - e. 3 miles
11. The condition known as Minamata disease was found to be caused by which one of the following pollutants?
 - a. Bacterial contamination
 - b. Methyl mercury *
 - c. Lead
 - d. Viral contamination
 - e. DDT
12. DDT seriously reduced the populations of marine birds in which one of the following ways?
 - a. It interfered with calcium deposition in the eggshells. *
 - b. It killed only young birds.
 - c. It killed only adult birds.
 - d. It killed the food of the birds.
 - e. It interfered with the birds' digestive process.
13. Presently, by international agreement, the following kinds of radioactive wastes are permitted to be dumped in the oceans:
 - a. Low-level wastes only *
 - b. High-level wastes only
 - c. All wastes
 - d. No wastes
14. Lost fishing gear continues to fish even though it is no longer being tended. This phenomenon is called _____.
 - a. ghost fishing *
 - b. lost fishing
 - c. empty fishing
 - d. nonfishing

15. Catching fish that you are not interested in is known as ____.
- bycatch *
 - nontarget catch
 - ghost fishing
 - empty fishing
 - lost fishing

True/False

Mark each of the given statements with the letter *T* if it is true and *F* if it is false.

[Instructors: In some cases, a few options are included for the same question.]

- [True] The major fisheries of the world are concentrated in waters overlying continental shelves.
- [False] The major fisheries of the world are concentrated in open oceanic waters.
- [True] Japan is a leading country in oyster mariculture.
- [False] The United States is a leading country in oyster mariculture.
- [True] Taiwan is a leading country in milkfish mariculture.
- [True] The distinction (cut-off value) between low- and high-level radioactive wastes is set at 1×10^{-8} curies per gram.
- [True] The author of the novel *Cannery Row* is John Steinbeck.
- [False] Algae is an important food for baleen whales.
- [False] Minamata disease is caused by lead poisoning.
- [False] The present fishing effort is divided almost evenly across the world's oceans.
- [True] The Pelican jumped into the water just for the hal-i-but.
- [True] Fishing may alter the species structure of the fished communities.
- [False] Capture of fishes for the aquarium trade does not produce significant losses.
- [True] Discharge of human sewage into the marine environment in small amounts and with adequate diffusing seems to cause no effects on marine communities on the open coast.

Short Answer (Fill in the Blank)

- Shell mounds excavated by archaeologists indicate that fisheries have been used for millennia.
- Most clupeoid fishes are harvested and processed for use as protein meal.
- Gadoids are a group of bottom-dwelling fishes including cod, haddock, and hake.
- The Peruvian anchoveta, *Engraulis ringens*, is eaten by several marine birds whose droppings, called guano, are harvested for fertilizer.
- The current of warm water that blocks upwelling along the Peruvian coast is called El Niño.
- A new fishery currently under development is based on small shrimplike animals called krill.
- The marine counterpart of agriculture is termed mariculture.
- The phenomenon of biological magnification is seen when the concentration of a chemical increases at each sequential trophic level.
- Although making a comeback now, the osprey was nearly extirpated from Florida by DDT contamination.
- Perturbations of the marine environment caused by the discharge of heat are called thermal pollution.
- Fishes represent 10% of the total protein intake by humans.
- The maximum amount of food that we can expect to harvest from the world's oceans each year without depleting their stocks is called the maximum sustainable yield.

184 Test Bank for Marine Biology

13. In 2001, the total marine fisheries landings were 83.6 million metric tons.
14. Fish currently being considered for increased harvest include southern ocean krill and squid, and the area of the continental slope at 300–1,000 meters is being considered as a new fishery.

Essay Discussion Questions

Develop complete answers to these questions.

1. Discuss how DDT has ended up in top marine carnivores.
2. Why is the introduction of alien animals into the marine environment of concern to marine biologists?
3. Present the crux (essence or core) of the argument known as the tragedy of the commons.
4. What did Garrett Hardin mean by the phrase “tragedy of the Commons”? How are the world’s oceans really commons? Discuss this concept, using the example of the *Exxon Valdez* oil spill in Alaska, the dumping of sewage and garbage by New York City, or the disposal of groundwater in your home state.
5. Who is Garrett Hardin, and what did he do that was so important for the ecological–environmental movement?
6. What is El Niño, and how does it affect the catch of the Peruvian anchoveta, *Engraulis ringens*? How does El Niño seem to influence weather patterns elsewhere in the world? Discuss.
7. What was the Cod War? How far out is the boundary for international waters?
8. What organisms are currently undergoing intensive mariculture or aquaculture practice? What are the drawbacks to such practice? What countries are heavily involved in aquaculture research? Why are these countries expending so much effort on this research?
9. What are the major pollution problems facing marine ecosystems? What are the major forces causing these problems?
10. What are the problems associated with oil spills? Consider the 1989 oil spill in Alaskan waters by the *Exxon Valdez*.
11. What are the problems associated with getting rid of sewage by dumping it into a bay?
12. What is meant by the terms “biological magnification” and “bioaccumulation”?
13. What happened on the shores of Minamata Bay? Describe the events of the tragedy. How did the methyl mercury get into the children?
14. What are DDT and DDE? How does DDT get into birds if they do not come anywhere near the places where the DDT is used?
15. Describe the fate of the Peruvian anchovy fishery from the mid-1950s to the present.
16. Why did the total catch in marine fisheries decline in the 1990s even though new resources such as krill were being exploited? What might this trend mean for the future of marine fisheries? Discuss.
17. Compare the fate of the Peruvian anchovy fishery from the mid-1950s to the present with that of the Great Lakes fishery from the mid-1800s to the present.
18. Speculate as to why the collapse of the Peruvian anchovy fishery in the mid-1950s had a serious impact on the price of chicken eggs in Europe.
19. Speculate as to what might have happened to the abundance of pelican eggs in Peru as a consequence of the collapse of the Peruvian anchovy fishery in the mid-1950s.
20. How does dredging influence productivity in the marine habitat? Discuss.
21. How can pet shops have an indirect impact on coral reefs?
22. Why does biological magnification work so well when the toxic agent is fat soluble?
23. Besides reducing the numbers of fishes, in what ways does fishing alter fish populations? Discuss.
24. What is the dead zone, and how do such places develop? Is there a natural counterpart to the dead zone? Discuss.

25. Of what importance is host shifting in disease-causing organisms in the marine environment?
26. Discuss how the explosion of technology has had significant effects on the ecology of the world's oceans in recent years.
27. Discuss how the rapidly increasing human population has had significant effects on the ecology of the world's oceans in recent years.
28. What is ghost fishing, and how does it effect the world's fisheries?
29. Do the decline of the Peruvian anchoveta fishery, the demise of the whale stocks, and the failure of the California sardine fishery have any driving forces in common? Discuss.
30. Challenge Question: In addition to depleting the stock of the target species, overfishing may result in changes in the larger ecosystems or communities in which the target species reside. Discuss how the effects of overfishing relate to the concept of ecosystem services.
31. Challenge Question: Support or refute the following statement: Only a whole-ecosystem approach can be useful to ensure a sustainable yield of the world's fisheries.
32. Challenge Question: Large amounts of human garbage are dumped into the ocean each year. Speculate on how this stuff might affect the ecosystem services of the world's oceans.
33. Challenge Question: When a dam is built on freshwater rivers, it can end up affecting the ocean. Discuss how this process occurs.
34. Challenge Question: How could a new sea-level Panama Canal potentially increase the numbers of alien species in the Pacific Ocean or Caribbean Sea? Discuss.
35. Challenge Question: Support or refute the following statement: Mariculture will solve the problem of feeding the human race. Provide specific examples to support your argument.
36. Challenge Question: How do pollutants reach the sea?
37. Challenge Question: How can those plastic-ring carriers that hold six-packs of beer and soda pop be an environmental problem?
38. Challenge Question: Discuss the concept of emerging disease in marine systems.
39. Challenge Question: Explore how you might be responsible for the degradation of the marine environment, if only in a minute way.
40. Challenge Question (from Economics 101): Many cities around the world discharge their untreated sewage directly into the marine environment. What will these cities have to do to internalize these externalities? Provide specific examples. *[Instructors: This question really will take an understanding of the economics concept of internalizing externalities.]*
41. Challenge Question: How might fecal pellets aid in the removal of certain radioactive materials from the oceans? Discuss.
42. Challenge Question: Compare the old paradigm and the new emerging appreciation of what oil spills do to marine ecosystems.
43. Challenge Question: How does the concept of ecosystem services relate to human impact on the sea? Discuss.

